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Pr. 2.25.

a) $uv^T = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix} [v_1 \dots v_n] =$ ~~$\begin{bmatrix} u_1v_1 & u_1v_2 & \dots & u_1v_n \\ u_2v_1 & \ddots & & u_2v_n \\ u_3v_1 & u_jv_j & & \vdots \\ \vdots & & & \\ u_nv_1 & \dots & \dots & u_nv_n \end{bmatrix}$~~

$= \begin{bmatrix} u_1v_1 & u_1v_2 & \dots & u_1v_n \\ u_2v_1 & u_2v_2 & \dots & u_2v_n \\ \vdots & & & \vdots \\ u_nv_1 & u_nv_2 & \dots & u_nv_n \end{bmatrix}$

~~we can see that rows 2, 3, ..., n~~

~~are~~ we can see that row j is just $\frac{u_j}{u_1}$ of the first row.

So, ~~applying~~

$R_j \leftarrow R_j - \frac{u_i}{u_1} R_1$ $(j \neq 1)$

to all the rows except R_1, we get.

$\begin{bmatrix} u_1v_1 & \dots & u_nv_n \\ 0 & 0 & 0 \end{bmatrix}$

that is why Rank = 1

b) ~~dim(~~

rank(A) = 1 means.

dim(row(A)) = 1. so, every row of A is just. multiple of some vector u which ~~is base~~ is a basis vector for row(A)

$A = \begin{bmatrix} c_1u \\ \vdots \\ c_ju \\ \vdots \\ c_nu. \end{bmatrix} = r_j(A) = c_ju$

$= \begin{bmatrix} c_1 \\ \vdots \\ c_j \\ \vdots \\ c_n \end{bmatrix} [u] = \begin{bmatrix} c_1 \\ \vdots \\ c_j \\ \vdots \\ c_n \end{bmatrix} [u_1 \dots u_i \dots u_n] = cu^T$

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Pr. 2.28

$$(A-UV^T)^{-1} = A^{-1} + A^{-1}U(1-V^TA^{-1}U)^{-1}V^TA^{-1}.$$

$$(A-UV^T)(A-UV^T)^{-1} = I = \cancel{I + U}$$

$$I = I + U(1-V^TA^{-1}U)^{-1}V^TA^{-1} - UV^TA^{-1} - UV^TA^{-1}U(1-V^TA^{-1}U)^{-1}V^TA^{-1}$$

$$U(1-V^TA^{-1}U)^{-1}V^TA^{-1} = UV^TA^{-1} + UV^TA^{-1}U(1-V^TA^{-1}U)^{-1}V^TA^{-1}$$

multiply both sides by A from right.

$$U(1-V^TA^{-1}U)^{-1}V^T = UV^T + UV^TA^{-1}U(1-V^TA^{-1}U)^{-1}V^T$$

$$U\left((1-V^TA^{-1}U)^{-1} - I - V^TA^{-1}U(I-V^TA^{-1}U)^{-1}\right)V^T = 0$$

$$\cancel{U((1-V^TA^{-1}U)^{-1}(I-V^TA^{-1}U).}$$

$$U\left((I-V^TA^{-1}U)(I-V^TA^{-1}U)^{-1} - I\right)V^T = 0$$

$$U\,0\,V^T = 0$$

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Pr. 2.34.

$x^T A x > 0$ for all $x \neq 0$

a) ~~$AA^{-1} = A^{-1}A = I$~~ $\det(A) \neq 0$

let's assume opposite if A is singular, then, there is a nonzero x that.

$Ax = 0$ → because $\det(A) = 0$ if A is singular

$x \neq 0$.

this means,

$x^T A x = 0$ for $x \in null(A)$ → which is not empty.

as this contradicts A being positive definite,

$\det(A) \neq 0$

b) $x^T A^{-1} x > 0$ → prove!

~~we can find $Ay = x$~~

~~$y^T A^T A^{-1} Ay > 0$~~

define y as:

if we ~~say~~ $Ay = x$

$x^T A^{-1} x = y^T A^T A^{-1} A y =$

$= y^T A^T y =$ because, this is a scalar value, taking the transpose results in the same scalar.

$= y^T A y$

because, A is one to one and onto,

there is a unique y for every x

and $x = 0$ iff $y = 0$.

so, when $y^T A y > 0$ is true,

$x^T A^{-1} x > 0$ is also true.

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Pr. 2.32.

a) $\|A\|_{max} = \max_{i,j} |a_{ij}|$

3 properties:

1) $\|A\| > 0$ $\|A\| = 0$ iff $A = 0$

2) $\|cA\| = |c| \|A\|$ for any scalar c

3) $\|A\| + \|B\| \geq \|A + B\|$

$|a_{ij}| \geq 0$ so, $\max_{i,j} |a_{ij}| \geq 0$

$\|A\| > 0$

$\|A\| = 0 \rightarrow \max_{i,j} |a_{ij}| = 0$

$0 = \max_{i,j} |a_{ij}| \geq a_{ij} \geq 0$

$0 \geq a_{ij} \geq 0$ so $a_{ij} = 0$ for all i,j

$\|cA\|_{max} = \max_{i,j} |c a_{ij}| = |c| \max_{i,j} |a_{ij}| = c \|A\|$

$\|A\|_{max} + \|B\|_{max} \geq \|A + B\|_{max}$.

$\max_{i,j} |a_{ij}| + \max_{i,j} |b_{ij}| \geq \max_{i,j} |a_{ij} + b_{ij}|$

for any i,j.

$\max_{i,j} |a_{ij}| + \max_{i,j} |b_{ij}| \geq |a_{ij}| + |b_{ij}| \geq |a_{ij} + b_{ij}|$

if we choose i and j such that.

$|a_{ij} + b_{ij}| = \max_{i,j} |a_{ij} + b_{ij}|$

we can say.

$\max_{i,j} |a_{ij}| + \max_{i,j} |b_{ij}| \geq \max_{i,j} |a_{ij} + b_{ij}|$

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8) $\|A\|_F = \left(\sum_{i,j} |a_{ij}|^2 \right)^{\frac{1}{2}}$

~~$\|A\| > 0$~~ $\sum_{i,j} \cancel{|a_{ij}|^2}$

$|a_{ij}| \geq 0 \rightarrow |a_{ij}|^2 \geq 0 \rightarrow \sum_{i,j} |a_{ij}|^2 \geq 0 \rightarrow \left(\sum_{i,j} |a_{ij}|^2 \right)^{\frac{1}{2}} \geq 0$

if $\|A\| = 0$. $\left(\sum_{i,j} |a_{ij}|^2 \right)^{\frac{1}{2}} = 0$

$\sum_{i,j} |a_{ij}|^2 = 0$

because $|a_{ij}| \geq 0$, we need every $|a_{ij}|^2 = 0$

$|a_{ij}| = 0$

$A = 0$

if all ~~$a_{ij} = 0$~~

$A = 0$, $a_{ij} = 0$.

all ~~a_{ij}~~ $\left(\sum_{i,j} |a_{ij}|^2 \right)^{\frac{1}{2}} = 0 \rightarrow \|A\| = 0$

$\|cA\|_F = \left(\sum_{i,j} |c a_{ij}|^2 \right)^{\frac{1}{2}} = \left(\sum_{i,j} |c|^2 |a_{ij}|^2 \right)^{\frac{1}{2}} = |c| \left(\sum_{i,j} |a_{ij}|^2 \right)^{\frac{1}{2}} = |c| \, \|A\|_F$

$\|a\|\|b\| \geq \|a \cdot b\|$

$\|A\|_F + \|B\|_F \geq \|A+B\|_F$

~~$|x|+|y| =$~~ Cauchy-Schwartz.

$\left(\sum_{i,j} |a_{ij}|^2 \right)^{\frac{1}{2}} + \left(\sum_{i,j} |b_{ij}|^2 \right)^{\frac{1}{2}} \geq \left(\sum_{i,j} |(a_{ij}+b_{ij})|^2 \right)^{\frac{1}{2}}$.

taking the square of both sides.

$\sum |a_{ij}+b_{ij}|^2 = \sum |a_{ij}|^2 + \sum |b_{ij}|^2 + 2\sum |a_{ij}||b_{ij}|$

$\left(\sum_{i,j} |a_{ij}|^2 \right) + \left(\sum_{i,j} |b_{ij}|^2 \right) + 2\left(\left(\sum |a_{ij}|^2 \right)\left(\sum |b_{ij}|^2 \right) \right)^{\frac{1}{2}} \geq \left(\sum_{i,j} |a_{ij}+b_{ij}|^2 \right)$

we subtract $\sum |a_{ij}|^2$ and $\sum |b_{ij}|^2$ from each side.

$\cancel{2}\left(\left(\sum |a_{ij}|^2 \right)\left(\sum |b_{ij}|^2 \right) \right)^{\frac{1}{2}} \geq \cancel{2} \sum |a_{ij}||b_{ij}|$

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Pr. 2.32.

b)

$$\cancel{\left(\sum |a_{ij}|^2\right)\left(\sum |b_{ij}|^2\right)}$$

this is cauchy - schwartz,

$$\left(\sum |a_{ij}|^2\right)\left(\sum |b_{ij}|^2\right)^{\frac{1}{2}} \geq \sum |a_{ij}||b_{ij}|$$ so, $\|A\|_F + \|B\|_F \geq \|A+B\|_F$.

~~Nijat~~

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a) It is more stable than normal Gaussian elimination because it ~~permutes one time~~ checks and if necessary, permutes rows one time. But after we are done with the first column and pass to the second, in the second column, it is possible that column entries below diagonal entry is bigger. In that case performing the algorithm will result in less stable algorithm than pivoting Gaussian

Answer: NO

b)

$$\begin{bmatrix} 1 & 2 & 3 \\ \varepsilon^2 & \varepsilon & 1 \\ 2 & 1 & 5 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

this is wrong because should hold

$2x_1 + x_2 + 5x_3 = 3$

$-10 + 10 = 0 \neq 3.$

$\frac{2}{\varepsilon - 2\varepsilon^2} - \frac{2}{\varepsilon - 2\varepsilon^2}.$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & \varepsilon - 2\varepsilon^2 & 1 - 3\varepsilon^2 \\ 0 & -3 & -1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 - \varepsilon^2 \\ 1 \end{bmatrix}$$

$x_3 = 2$

$x_2 = 0$

$x_1 = -5$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & \frac{1 - 3\varepsilon^2}{\varepsilon - 2\varepsilon^2} \\ 0 & 0 & \frac{3 - 9\varepsilon^2}{\varepsilon - 2\varepsilon^2} - 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ \frac{2 - \varepsilon^2}{\varepsilon - 2\varepsilon^2} \\ \end{bmatrix}$$

after app. ...

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & \frac{1 - 3\varepsilon^2}{\varepsilon - 2\varepsilon^2} \\ 0 & 0 & \frac{3 - \varepsilon - 7\varepsilon^2}{\varepsilon - 2\varepsilon^2} \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ \frac{2 - \varepsilon^2}{\varepsilon - 2\varepsilon^2} \\ 1 + 3\frac{2 - \varepsilon^2}{\varepsilon - 2\varepsilon^2} \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & \frac{1}{\varepsilon - 2\varepsilon^2} \\ 0 & 0 & \frac{3}{\varepsilon - 2\varepsilon^2} \; 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ \frac{2}{\varepsilon - 2\varepsilon^2} \\ \end{bmatrix}$$